

Nanyang JC J2 Preliminary Exam 2018
H1 Chemistry 8873/01
Paper 1 MCQ Answers and Comments

Qn	Ans	Qn	Ans	Qn	Ans	Qn	Ans	Qn	Ans	Qn	Ans
1	B	6	C	11	C	16	B	21	B	26	A
2	B	7	A	12	C	17	A	22	A	27	C
3	C	8	B	13	D	18	D	23	D	28	B
4	D	9	D	14	C	19	C	24	B	29	D
5	A	10	C	15	C	20	B	25	D	30	B

1 B

Ar of A

$$= \frac{(140 \times 2) + (142 \times 3) + (144 \times 1)}{2 + 3 + 1} = \frac{850}{6}$$

$$= 141.67 = \underline{141.7} \text{ (1 d.p)}$$

2 B

$2 \text{ Cl} \equiv \text{Cl}_2$

$$\text{Amount of Cl atom} = \frac{1}{24\,000} \times 2 \text{ mol}$$

$$\text{No. of Cl atom} = \text{amt} \times \text{Avogadro constant}$$

$$= \frac{1 \times 6.02 \times 10^{23} \times 2}{24\,000}$$

3 C

Cr is oxidised from +3 to +6

Fe is oxidised from +2 to +3

Oxygen is reduced from 0 to -2

4 D

Cu: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$

Fe^{3+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

Ni^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$

Sr^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

5 A

Cations are smaller than their respective atoms as cations have **one electron shell less** than neutral atoms.

Anions are bigger than their respective atoms because of the extra repulsion of the added electron(s). Since the nuclear charge **remains the same**, the outermost electrons are now less attracted to the nucleus.

Anions are bigger than cations because anions contain **one more electron shell** and the increase in screening effect outweighs the increase in nuclear charge.

Atomic radius decreases across the period.

6 C

A BrF_3 : T-shaped (3bp, 2 lp)

B ICl_4^- : square planar (4bp, 2 lp)

C PCl_4^+ : tetrahedral (4 bp, 0 lp)

D XeF_4 : square planar (4bp, 2 lp)

7 A (1 & 4 are correct)

1 True, bond angle in N_2H_4 is 107° while bond angle in H_2O_2 is 105°

2 False, the N–H bond is *longer* than the O–H bond as the O–H covalent bond is more polar. Students can also check the DB to compare the 2 bond energies.

3 False, hydrazine forms *weaker* intermolecular hydrogen bonds than hydrogen peroxide as N is less electronegative than O.

4 True, both hydrazine and hydrogen peroxide are polar molecules. In hydrazine, the shape is trigonal pyramidal shape about N, while in hydrogen peroxide, the shape is bent shape about O. The dipoles do not cancel out.

8 B

E: must be a liquid at room temperature as its mpt < room temp < boiling pt. Hence E must be Br₂. Options C and D are eliminated.

F & G have relatively low melting points, hence they are unlikely to be Al and Si. Si having a giant covalent structure will have a mpt > 1000°C. Option A is eliminated.

Option B is the best answer. Between Na and Mg, Na will have lower melting point and boiling point as Na has weaker metallic bonding due to lesser delocalised electrons available. Na will have lower density (mass divided by volume) as ionic radius of Na⁺ is larger than that of Mg²⁺ and mass of Na⁺ and Mg²⁺ is approximately constant.

9 D

The activation energy for the forward reaction is smaller than that for the backward reaction.

10 C

$$\text{amount of water formed} = \frac{25.0}{1000} \times 3.00 = 0.07500 \text{ mol}$$

$$\begin{aligned} \text{heat produced} &= (25.0 + 25.0) \times 4.18 \times 9.5 \times \frac{100}{85} \\ &= 2335 \text{ J mol}^{-1} \end{aligned}$$

$$\begin{aligned} \Delta H_{\text{r}} &= -\frac{2335 \times 10^{-3}}{0.07500} \\ &= -31.14 \\ &= -31.1 \text{ kJ mol}^{-1} \end{aligned}$$

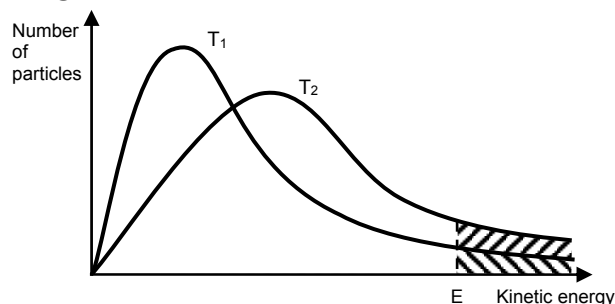
11 C

$$\begin{aligned} \Delta H_{\text{r}}^{\circ} \text{ lattice energy CaO} &= -(\Delta H_{\text{r}}^{\circ} \text{ atomisation Ca} + 1^{\text{st}} \text{ and } 2^{\text{nd}} \text{ ionisation energies of calcium} + 950) \\ &+ \Delta H_{\text{r}}^{\circ} \text{ formation CaO} \end{aligned}$$

Option A: Less exothermic
Option B: Less exothermic

Option C: More exothermic as 1st and 2nd ionisation energies of magnesium are more endothermic than calcium

Option D: Standard enthalpy change of combustion of magnesium is equal to the standard enthalpy change of formation of magnesium oxide

12 C

Option 1 and 2 are correct. See peak of T2
Option 3 is correct. The area under the curve increases for energy greater than E.
Option 4 is wrong. The proportion of molecules at lower energies decreases.

13 D

At high ethanal concentration all the active sites in the enzyme molecules are occupied by ethanal molecules, hence the rate does not increase even though [ethanal] concentration increases i.e. zero order kinetics.

14 C

$$\begin{aligned} [Q]_{\text{eqm}} &= 0.2 / 10 \text{ mol dm}^{-3} \\ [PQ_3]_{\text{eqm}} &= 0.5 / 10 \text{ mol dm}^{-3} \\ \text{Initial P} &= 0.2 \text{ mol} \end{aligned}$$

Since P and PQ₃ have the same mole ratio in the equation,
|change in P| = |change in PQ₃| = 0.1
(read off from graph)

Alternatively, you can also find change in P = 1/3 change in Q.
Hence [P]_{eqm} = (0.2 – 0.1) = 0.1 / 10 mol dm⁻³

$$K_c = \frac{[PQ_3]}{[Q]^3[P]} = \frac{\left(\frac{0.5}{10}\right)}{\left(\frac{0.2}{10}\right)^3 \left(\frac{0.1}{10}\right)} = \frac{0.5 \times 10^3}{0.2^3 \times 0.1}$$

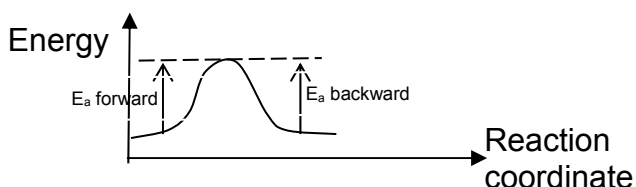
15 C

If equilibrium constant is independent of temperature, it means when temperature changes, neither forward nor backward reaction is favoured and this is only possible if ΔH is zero.

1. All rate constants are affected by temperature according to Arrhenius equation: $k = A^{-E_a/RT}$.

2. Temperature would have no effect on the position of equilibrium if ΔH is zero.

3. When ΔH is zero, activation energies for forward and backward reaction are equal as shown in the energy profile diagram below:



4. We cannot deduce any information about the number of gaseous particles unless some information about how value of equilibrium constant changes with pressure is provided.

16 B

A. A weak acid is partially dissociated in aqueous solution. A weak acid has a low K_a value hence it has a high pK_a value.

B. A weak acid has a strong conjugate base. Hence statement B is incorrect.

C. Acids are proton donors.

D. There is only a low concentration of ions formed as the weak acid remains mostly undissociated (as a molecule). Hence, there are few mobile ions to act as charge carriers, accounting for the low electrical conductivity.

17 A

A buffer contains a weak acid or base and its conjugate salt hence two possible buffers can be formed from HCO_3^- :

H_2CO_3 (weak acid) and HCO_3^- (conj base)

HCO_3^- (weak acid) and CO_3^{2-} (conj base)

Hence options 1 and 2 are correct.

For option 3:

$n(\text{H}^+) = 2 \times n(\text{H}_2\text{SO}_4) = n(\text{HCO}_3^-)$.

H_2SO_4 reacts completely with HCO_3^- to form H_2CO_3 . No excess HCO_3^- is left hence a buffer is not formed.

18 D

Properties of:

MgO – basic oxide. Slightly soluble in water. It reacts with acid to form salt and water.

SiO_2 – insoluble in water and unreactive to dilute acids and bases.

MgCl_2 – soluble in water to give a weakly acidic solution.

Al_2O_3 – insoluble in water. Amphoteric in nature and reacts with both acids and bases to give salt and water.

I: Add water to the mixture. Stir well and filter the mixture.

Residue contains: MgO , Al_2O_3 , SiO_2

Filtrate contains: MgCl_2 , MgO as $\text{Mg}(\text{OH})_2$

II: To the residue, add $\text{NaOH}(\text{aq})$ and stir well with a glass rod.

Only Al_2O_3 reacts with $\text{NaOH}(\text{aq})$

$\text{Al}_2\text{O}_3(\text{s}) + 2\text{OH}^-(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow 2[\text{Al}(\text{OH})_4]^- (\text{aq})$

III: Filter the resulting mixture.

Residue contains: SiO_2 , MgO

Filtrate contains: $[\text{Al}(\text{OH})_4]^- (\text{aq})$

IV: To the residue, add $\text{HCl}(\text{aq})$ and stir well with a glass rod.

Only MgO reacts with HCl .

$\text{MgO}(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

V: Filter to obtain the filtrate and residue.

Residue contains: SiO_2

Filtrate contains: $\text{Mg}(\text{OH})_2(\text{aq})$

1 – No. Both SiO_2 and MgCl_2 remained unreacted. MgCl_2 only dissolved in water in step I therefore remained unreacted.

- 2 – No. Filtrate contains both $\text{Mg}(\text{OH})_2$ and MgCl_2 .
 3 – No. The residue contains SiO_2 and MgO .
 4 – No gas was produced throughout the separation procedure.

19 C

A – From Na_2O to Al_2O_3 , the covalent character increases due to the increasingly higher charge density of the cation.

$$\mathbf{B} - |\text{LE}| \propto \frac{|z_+ z_-|}{r_+ + r_-}$$

When the anion is large, the changes in the sum of ionic radii becomes negligible. The product of ionic charges increases from sodium to aluminium, thus lattice energy become more exothermic.

C – The oxides become less soluble across the Period from sodium to silicon. (See Periodicity notes – Reaction of oxides with water)

D – Melting points increases from sodium to silicon and decreases from silicon to argon. (Giant metallic structure $\rightarrow$ Giant molecular structure $\rightarrow$ simple molecular structure) (See Periodicity notes for full explanations on melting points)

20 B

The chloride of X react with water to give a colourless acidic solution.

The chlorides which are acidic in water are MgCl_2 , AlCl_3 , SiCl_4 and PCl_5 .

SiCl_4 reacts with water to form a solid SiO_2 .

$\Rightarrow$ X can be Mg, Al or P.

The oxides of X and Y are insoluble in water.

Only 2 oxides in Period 3 are insoluble in water, Al or Si.

$\Rightarrow$ X must be Al (agree with first point)

$\Rightarrow$ Y must be Si

Oxides of X and Y have high melting and boiling points. Oxide of Z has low melting and boiling points.

Across Period 3, the oxides change from giant ionic structures to giant covalent structures to simple molecular structures.

$\Rightarrow$ Element Z has a simple molecular structure, so it is either phosphorus, sulfur or chlorine.

$\Rightarrow$ Element Z forms anions and has the largest ionic radius.

Order of increasing ionic radius:

Si^{4+} , Al^{3+} , P^{3-} (or S^{2-} or Cl^-)

21 B

An unsaturated 5-carbon alcohol would have molecular formula, $\text{C}_5\text{H}_{10}\text{O}$, e.g. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

E has 4 less H atoms which implies that there is presence of 2 multiple bonds which could be $\text{C}=\text{C}$ or $\text{C}=\text{O}$ bond.

Hence, the possible isomers are

$\text{CH}_2=\text{CHCH}=\text{CHCH}_2\text{OH}$,

$\text{CH}_2=\text{CHCH}_2\text{COCH}_3$ and

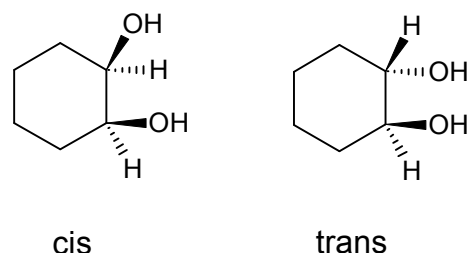
$\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{CHO}$.

Hence, there can be alkene, alcohol, ketone and aldehyde present in the isomers of $\text{C}_5\text{H}_8\text{O}$. (1 and 2 only)

An ester is not possible as there is only 1 oxygen atom present in E.

22 A

The product after addition reaction with H_2 is as follows:

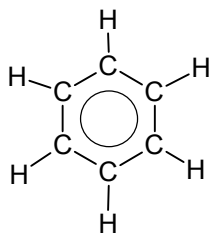


A:

B: It is an ester and esters do not react with H_2 .

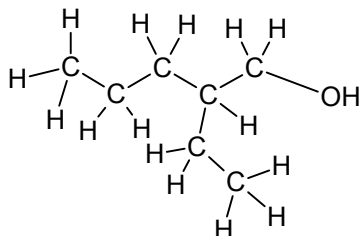
C: $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ do not exhibit cis-trans isomerism.

D: It is an amide and amides do not react with H_2 .

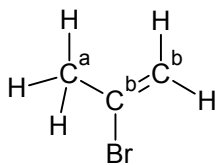
23 D

A: All of its carbons have 3 bond pairs and 0 lone pair. Bond angle: 120° .

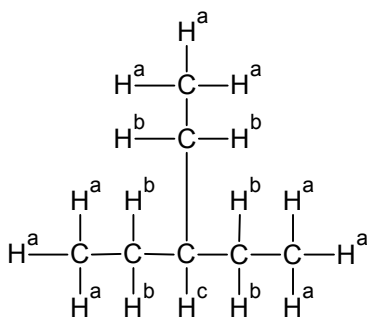
B: Its carbon have 3 bond pairs and 0 lone pair. Bond angle: 120° .



C: All of its carbons have 4 bond pairs and 0 lone pairs. Bond angle: 109.5° .



D: C^a have 4 bond pairs and 0 lone pair. Bond angle about C^a : 109.5° . Both C^b have 3 bond pairs and 0 lone pair. Bond angle about C^b : 120° .

24 B

The chlorine atom can substitute once with H^a , H^b or H^c . (i.e. there are 3 types of hydrogen atoms - H^a , H^b and H^c , with different chemical environments) Hence, there can only be 3 possible structural isomers of monochloro-compounds.

25 D

A: Both compounds are able to liberate Cl^- to give white ppt of $AgCl$ with ethanolic $AgNO_3$.

B: Solution remains colourless for both compounds.

C: Solution remains colourless for both compounds.

D: CH_3CH_2Cl will liberate Cl^- to give white ppt of $AgCl$ with ethanolic $AgNO_3$ whereas CH_3CH_2Br will liberate Br^- to give cream ppt of $AgBr$ with ethanolic $AgNO_3$.

26 A

A: CH_3COCH_3 will not get oxidised or undergo acidic hydrolysis.

B: CH_3CH_2CHO will get oxidised to CH_3CH_2COOH (propanoic acid).

C: $CH_3CH_2CH_2OH$ will get oxidised to CH_3CH_2COOH (propanoic acid).

D: $CH_3CH_2CONHCH_3$ will undergo acidic hydrolysis to give CH_3CH_2COOH (propanoic acid) and $CH_3NH_3^+$.

27 C

A: The non-polar regions of the polymer is too large to be dissolved in water.

B: PP has a giant molecular structure such that the non-polar oil molecules are unable to form favourable intramolecular forces of attraction to provide sufficient energy to overcome the extensive intermolecular forces of attraction within the PP polymer.

C: PP is unable to undergo alkaline hydrolysis but the ester groups in PET can undergo strong alkaline hydrolysis.

D: Both are able to store hot foods as they have extensive intramolecular forces of attraction that are strong enough to withstand such heat.

28 B

- 1: Polyesters are condensation polymers.
2: Poly(ethene) are addition polymers.
3: poly(vinyl alcohol) are addition polymers.
(2 & 3 only)

29 D

It is the low porosity of the carbon nanotubes that made them to be a suitable material for water filtration. The size of the hexagonal 'holes' in carbon nanotubes are small enough to allow water molecules to pass through but not for other larger species (e.g. other ions, particles etc).

30 B

A: Their health risks are poorly understood. They may pose potential health risks.

B: Nanomaterials can have a length of more than 100 nm as they just need at least one dimension in the size of 1 to 100 nm.

C: Nanomaterials can be both man-made and exist biologically.

D: Not all nanomaterials can stick onto surfaces (e.g. catalytic converter).